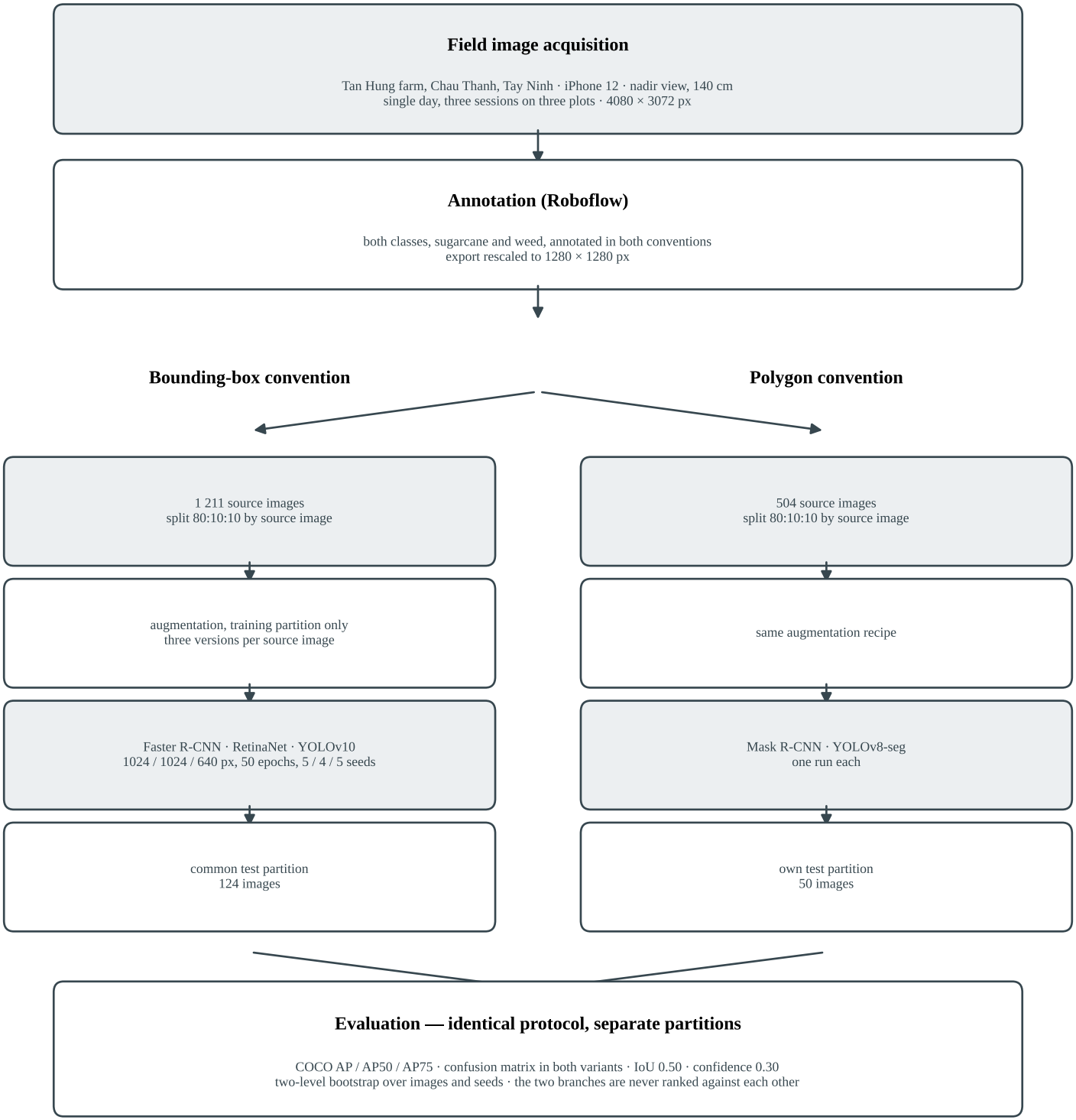
Field image acquisition
Tan Hung farm, Chau Thanh, Tay Ninh · iPhone 12 · nadir view, 140 cm
single day, three sessions on three plots · 4080 × 3072 px
Annotation (Roboflow)
both classes, sugarcane and weed, annotated in both conventions
export rescaled to 1280 × 1280 px
Bounding-box convention
Polygon convention
1 211 source images
split 80:10:10 by source image
504 source images
split 80:10:10 by source image
augmentation, training partition only
three versions per source image
same augmentation recipe
Faster R-CNN · RetinaNet · YOLOv10
1024 / 1024 / 640 px, 50 epochs, 5 / 4 / 5 seeds
Mask R-CNN · YOLOv8-seg
one run each
common test partition
124 images
own test partition
50 images
Evaluation — identical protocol, separate partitions
COCO AP / AP50 / AP75 · confusion matrix in both variants · IoU 0.50 · confidence 0.30
two-level bootstrap over images and seeds · the two branches are never ranked against each other